

Saurabh Chauhan

Curriculum Vitae

Computer Science student at NIT Hamirpur focused on backend engineering, distributed systems, AI-powered applications, and developer tooling.

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EDUCATION

National Institute of Technology Hamirpur

Bachelor of Technology in Computer Science and Engineering

Aug. 2024 – May 2028

CGPA: 9.16 / 10.0

TECHNICAL SKILLS

Programming Languages: C++, Go, JavaScript, SQL

Backend Development: Node.js, Express.js, RESTful APIs, gRPC, Microservices Architecture

Databases & Infrastructure: PostgreSQL, Redis, Docker, Git, Linux, Postman

Core Competencies: Data Structures and Algorithms, System Design, Distributed Systems, Concurrency

PROJECTS

GoCrawl – Concurrent Web Crawler 🐙 | Go, Goroutines, Software Engineering

Jun. 2026

- Developed a scalable web crawler in Go leveraging goroutines and channels with a **5-worker pool**, enabling parallel URL fetching across **500+ pages** per execution.
- Engineered thread-safe deduplication algorithms using `sync.RWMutex` and per-domain rate limiting, reducing redundant HTTP requests by **75%+**.
- Designed an HTML parsing and URL normalization pipeline with Go's `net/html` for robust link discovery and same-domain task scheduling.
- Implemented timeout protocols, redirect validation, and race condition detection, achieving **85%+** test coverage.

CryptX – Parallel File Encryption System 🐙 | C++23, POSIX, Multi-processing

Jun. 2026

- Engineered a parallel file encryption engine in C++23 capable of recursively encrypting and decrypting directory trees through low-level binary I/O operations.
- Established a POSIX `fork()/exec()` architecture with task queues, facilitating **concurrent cryptographic operations** across isolated child processes.
- Designed a highly modular software architecture featuring task serialization, I/O abstractions, and environment-based secure key management.

ApeAI – Automated Product Engineering AI 🐙 | Node.js, Supabase, pgvector, Agentic AI

May 2026

- Developed a **5-layer, fully decoupled** product operations platform using **Node.js** that converting multi-source customer feedback into structured Jira/GitHub engineering tickets with **zero manual re-entry**.
- Engineered an AI pipeline using **Google Gemini** to generate **768-dimensional embeddings** and cluster signals via **cosine similarity** on **Supabase pgvector**, enabling automated semantic grouping of unstructured inputs.
- Automated generation of **BRDs, PRDs, Agile user stories, and technical tasks** using **structured JSON mode**, reducing pre-development workflow from **weeks to minutes**.
- Built a **universal formatter** to integrate with **Jira APIs**, auto-generating labelled, prioritized tickets and streamlining handoff from planning to execution.

GoLink – Distributed URL Shortener 🐙 | Go, PostgreSQL, Redis, Docker, gRPC

May 2026

- Architected a high-throughput distributed URL shortener utilizing Go, PostgreSQL, and gRPC; support custom aliases, TTL expiration, and IP-based rate limiting (**10 req/min**).
- Integrated a Redis cache-aside caching pattern, serving **~95%** of repeat queries from memory and reducing latency from **~10ms to ~2ms**.
- Deployed **AI-assisted slug generation** via LLMs to create context-aware and SEO-optimized short URLs.
- Containerized **4 microservices** using Docker Compose to ensure a seamless, zero-configuration deployment environment.

CodeSync – Real-Time Collaborative Code Editor 🐙 | Yjs, Liveblocks, CodeMirror

Mar. 2026

- Built a real-time collaborative code editor using **Next.js 16**, **React 19**, and **CodeMirror 6**, enabling simultaneous multi-user editing with instant synchronization.
- Implemented conflict-free collaboration using **Yjs CRDTs**, ensuring consistent document state across distributed clients without manual merge resolution.
- Integrated **Liveblocks** for live cursors, presence awareness, avatars, and per-user undo/redo history, delivering a Google Docs-style collaborative coding experience.

- Built a **non-custodial decentralized application** on Stellar enabling users to create rate-protected foreign exchange intents with configurable execution windows and atomic settlement guarantees.
- Integrated **Stellar Path Payments**, real-time orderbook analysis, and liquidity estimation to execute conversions only when market conditions satisfy user-defined exchange rates.
- Engineered wallet connectivity, transaction signing, trustline management, and transaction monitoring using the **Stellar SDK** and **Freightier Wallet API**.

- Built a full-stack resume generation platform using **React 19**, **Node.js**, and **Express.js**, enabling resume customization, section reordering, template selection, and multi-format exports.
- Integrated **Google Gemini 2.0 Flash** for ATS-focused content enhancement, automatically rewriting experience, projects, and skills into recruiter-friendly descriptions.
- Implemented a server-side **LaTeX compilation pipeline** using **pdflatex** for real-time preview and professional PDF generation.

LEADERSHIP & EXPERIENCE

Google Developer Groups Ludhiana – NIT Hamirpur Chapter

Jan. 2025 – Present

Executive

- Spearheaded the Genesis initiative, successfully onboarding **450+ students** and organizing **10+ technical workshops** focused on software engineering fundamentals.
- Directed GDG Abyss, coordinating technical challenges and algorithmic coding competitions for **300+ participants**.

TEDxNIT Hamirpur

Nov. 2025 – Present

Volunteer – Web Developer

- Developed and deployed the TEDxNIT Hamirpur 2026 website using Next.js and Vercel, boosting event registrations; site served **100+ unique visitors** post-launch. `</>`
- Managed speaker onboarding and scheduling for **7 speakers**, coordinating **15+ cross-functional communications** to ensure full TEDx guideline compliance.

Google Student Ambassador 2026

May 2026 – Present

Campus Ambassador | GID: 4262

- Selected to represent Google developer initiatives, managing technical outreach and program engagement for **50+ engineering students**.
- Hosted **3+ technical workshops** emphasizing Google developer tools, backend architecture, and AI/ML deployments.

ACHIEVEMENTS

- Ranked in the **top 3.8%** of the entire 2028 batch at NIT Hamirpur with a CGPA of **9.16/10.0**.
- Awarded a technical contribution certificate for developing and optimizing features for the official NIT Hamirpur website.